Java Practice Questions

1. Functions / Methods

1. Write a function to print your name.
2. Write a function to calculate the sum of two numbers.
3. Write a function to calculate the product of two numbers.
4. Write a function to calculate the factorial of a number.
5. Write a function to check whether a number is even or odd.
6. Write a function to print the multiplication table of a number.
7. Write a function to check whether a number is prime or not.
8. Write a function to check voting eligibility based on age.

2. Conditional Statements

9. Write a program to check whether a number is positive, negative, or zero.
10. Write a program to check whether a number is even or odd.
11. Write a program to find the greater of two numbers.
12. Write a program to find the greatest of three numbers.
13. Write a program to check whether a year is a leap year.
14. Write a program to assign grades according to marks:
    - 90–100 → A
    - 75–89 → B
    - 60–74 → C
    - 40–59 → D
    - Below 40 → Fail
    - Above 100 or below 0 → Invalid
15. Write a program to check whether a character is a vowel or consonant.
16. Write a program to calculate an electricity bill using slabs:
    - First 100 units → ₹5/unit
    - Next 100 units → ₹7/unit
    - Above 200 units → ₹10/unit
17. Write a program to categorize a person based on age:
    - Child
    - Teenager
    - Adult
18. Write a program to check whether a person is eligible to vote based on:
    - Age
    - Indian citizenship

3. Nested if

19. Write a program to check voting eligibility using nested if.
20. Find the greatest of three numbers using nested if.
21. Check whether a student passes based on:

- Marks ≥ 40
- Attendance ≥ 75%

4. Switch Statements

22. Create a program that takes a number from 1–7 and prints the corresponding day.
23. Create a calculator using switch for:
    - +
    - -
    - *
    - /
24. Create a program that takes a number from 1–12 and prints the corresponding month.
25. Check whether a day is a weekday or weekend using switch.
26. Create a food menu using switch:
    - 1 → Pizza
    - 2 → Burger
    - 3 → Sandwich
    - 4 → Pasta
    - 5 → Exit
27. Create an ATM menu using switch:
    - Check balance
    - Deposit
    - Withdraw
    - Exit
28. Create a traffic signal program using switch:
    - r → Stop
    - y → Get ready
    - g → Go

5. FOR Loops

29. Print numbers from 1 to 10.
30. Print numbers from 10 to 1.
31. Print all even numbers from 1 to 20.
32. Print all odd numbers from 1 to 20.
33. Print the multiplication table of a number.
34. Find the sum of numbers from 1 to 10.
35. Find the sum of even numbers from 1 to 20.
36. Count how many numbers are divisible by 5 from 1 to 50.
37. Find the factorial of a number using a for loop.

6. WHILE Loops & Number Problems

38. Reverse a number.
39. Find the sum of digits of a number.
40. Count the number of digits in a number.

41. Check whether a number is a palindrome.
42. Check whether a number is prime.
43. Print the Fibonacci series.
44. Find the largest digit in a number.
45. Find the smallest digit in a number.
46. Count the even and odd digits in a number.
47. Find the sum of even digits and sum of odd digits separately.
48. Find the product of all digits of a number.
49. Count how many times a particular digit occurs in a number.
50. Find the sum of numbers from 1 to n using a while loop.
51. Print all numbers from 1 to n that are divisible by 3.
52. Count the even and odd numbers from 1 to n.
53. Find the largest number among n numbers entered by the user.
54. Find the HCF/GCD of two numbers using the Euclidean Algorithm.
55. Check whether a number is a perfect number.